

Data Science and Machine Learning

Generative AI II

Retrieval augmented generation (RAG)

RAGs

RAG is a cutting-edge NLP technique that combines document retrieval with generative models. It retrieves relevant information from a database and uses it as input to generate accurate, context-specific response

Benefits:

- ✅ Accuracy: Responses are grounded in reliable sources.
- 🔷 Context Awareness: Adaptable to the user's input and specific needs.
- 🔷 Scalability: Handles vast amounts of data efficiently.Applications:
- 🔌 Chatbots for customer support.
- 👷 Question-answering systems for research.
- 📈 Technical document analysis for specialists

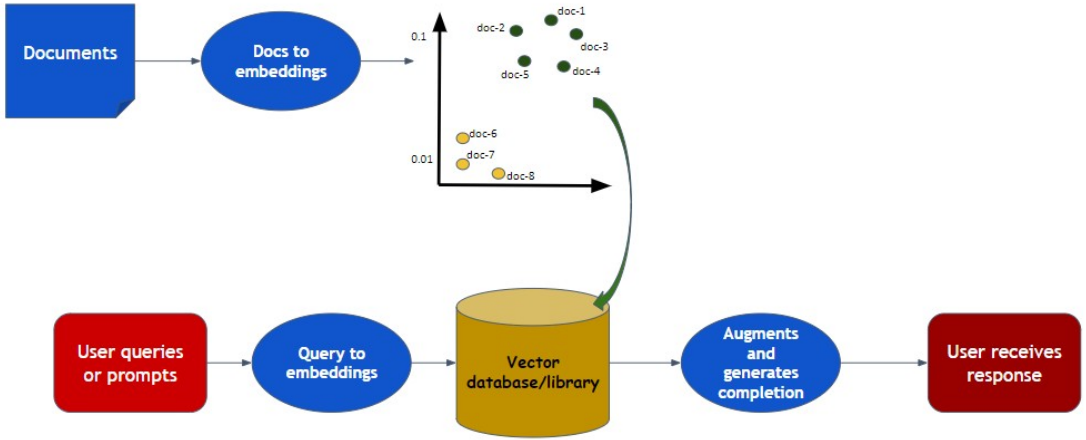
RAG vs Fine-Tuning

- **No training required** – You don’t need to modify the model.
- **Faster and cheaper** – No GPU training, no long processing times.
- **Easily updatable** – Just update or swap the documents, no need to retrain.
- **Access to recent or private data** – You can include documents with up-to-date or company-specific information.
- **Explainable outputs** – RAG retrieves relevant documents, so you can see where the answer came from.
- **Lower risk of overfitting** – Your data stays external; the model doesn't memorize it.
- **Scalable** – Add more documents without increasing the model size.

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RAG Workflow

- 🔎 **Document Retrieval**: Locate relevant information from a document store or database based on a user's query.
- 🔬 **Embeddings**: Use embeddings (dense numerical vectors) to represent text, enabling efficient similarity search and retrieval.
- 🤖 **Generative AI**: Feed the retrieved data into a generative model to produce responses tailored to the user's query.



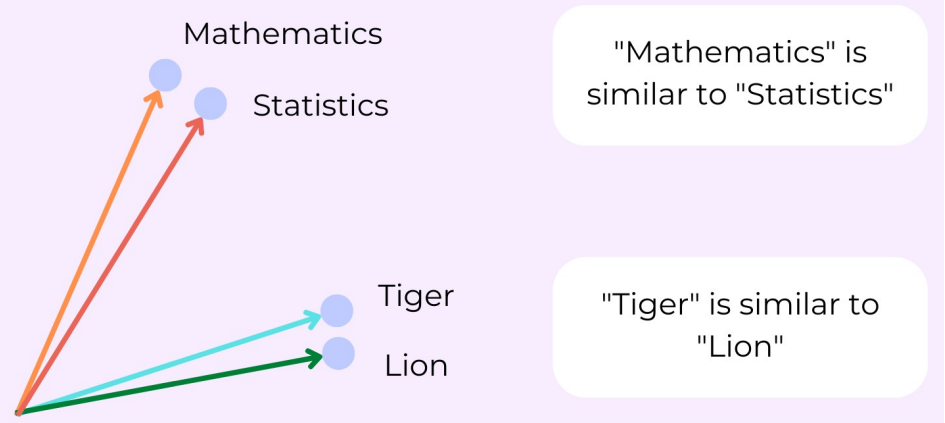
Embeddings Overview

🔷 **OpenAI Embeddings**: High-quality, general-purpose models (e.g., `text-embedding-ada-002`).

🔷 **SentenceTransformers**: Lightweight, domain-specific models like `all-MiniLM-L6-v2`.

Cosine Similarity

Cosine similarity measures how similar two vectors are by calculating the cosine of the angle between them.



ChromaDB

ChromaDB

ChromaDB is a vector database designed for efficiently storing and retrieving embeddings.

- **Why Use ChromaDB?**
  - Enables fast similarity searches.
  - Scales well with large datasets.
- **How It Works**: Documents ➔ Embeddings ➔ Stored in ChromaDB for efficient retrieval.

